BLOOD TISSUE

1 - Definition

- Blood (approximately 5L in adults) is a mesenchymal tissue, a fluid contained within the vascular system: a closed network of cavities (the blood vessels).
Consisting of a complex liquid, **plasma,** in which **blood cells** circulate in suspension.

It is a thick, red liquid, slightly alkaline (pH between 7.35 and 7.45), and plays an important role in homeostasis:

- It transports oxygen and nutrients to the body's cells;
- It removes waste products from cellular metabolism (carbon dioxide, nitrogenous waste) which will then be eliminated by the lungs, kidneys and sweat;
- It protects the body (white blood cells, antibodies, platelets);
- It maintains the hydro-electrolytic balance.

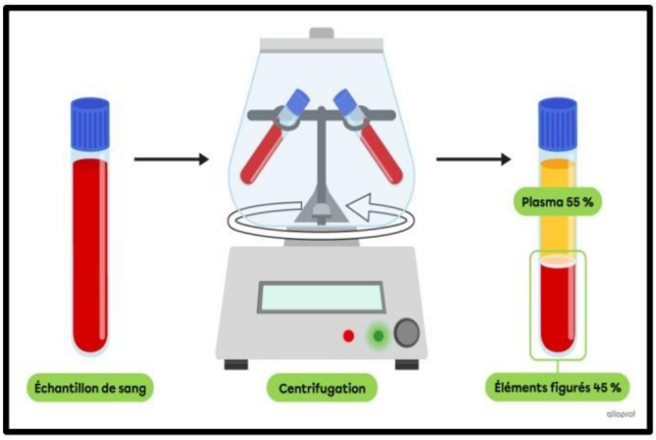
2- Plasma:

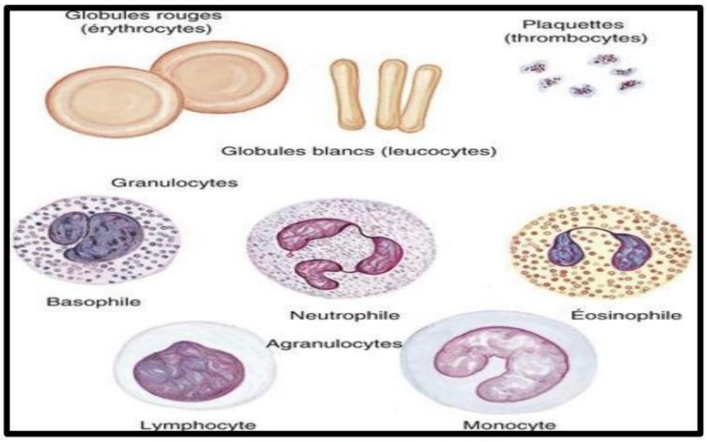
A- Structure:

Plasma is isolated by centrifugation of a blood sample taken over an anticoagulant.
Plasma is the liquid part of blood in which blood cells are suspended. It is straw-yellow in color and represents 55% of the blood volume.
Plasma is composed of 90% water, however it is a complex liquid that includes a significant number of organic or inorganic substances.
- The most abundant organic substances are proteins (7% of the total plasma weight), three main groups are distinguished: • albumins • globulins • fibrinogen, with glucose and lipids.

- Plasma also contains ions such as sodium, potassium, chloride, calcium, magnesium, bicarbonate and phosphate. It carries a small amount of gas, as well as hormones, enzymes and nutrients.

- Finally, it transports metabolic waste (urea, uric acid, etc.) and products resulting from the metabolism of hormones and drugs.





Blood centrifugation

Fig 1 Figurative elements of blood

B- Plasma fulfills several functions:

- The transport of blood cells and nutrients; the regulation of water and mineral salts in the body and the irrigation of tissues.
- The albumin contained in the plasma prevents the blood from losing too much water and thickening as it circulates through narrow vessels permeable to water (capillaries).

Albumin transports various blood components and nutrients. Furthermore, immunoglobulins in the plasma are antibodies that, along with white blood cells, play an important role in defending against pathogens.

- Then, coagulation factors, along with platelets, are responsible for stopping bleeding.
- A deficiency in these proteins can lead to an inability to retain water in the vessels (albumin), a decrease in the body's immune defenses (immunoglobulins), or abnormalities in blood clotting (coagulation factors).

3- Blood cells (formed elements of blood):

- Blood cells are:
- Red blood cells (or erythrocytes, or normocytes), white blood cells (or leukocytes) and platelets (or thrombocytes),
- Blood cells originate from a primitive cell (multipotent cell) in the bone marrow

bony.

- This is a soft substance contained in the epiphyses of long bones and flat bones.

3-1- Red blood cells, or erythrocytes, or normocytes:

- Red blood cells can be observed in the fresh state, between a slide and a coverslip or on a blood smear; after staining, they appear reddish-brown (Wright stain) or pink (May-Grünwald-Giemsa).

-The erythrocyte has a lifespan of 100 to 120 days. Physiological hemolysis then occurs – this is carried out by macrophages, phagocytic cells, found in the liver

, the spleen , bone marrow and lymph nodes.

- The normal concentration of RBCs in the blood is 3.9 to 5.5 million per mm3 in women and 4.1 to 6 million per mm3 of blood in men.
- These are highly differentiated cells that have lost their nucleus through a phenomenon of cell exclusion; they are terminal cells.
- These cells have a rounded, biconcave disc shape.
- The plasma membrane has a particular composition, which will contribute to maintaining the biconcave disc shape.
- This special structure will have a larger surface area than if the cell were rounded, and therefore a better exchange surface between the internal and external environments.
- They carry on their surface the antigens of the erythrocyte blood groups ABO, rhesus, Kell, ….

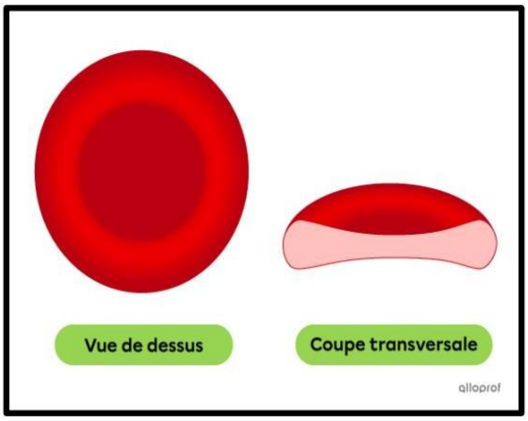
- The diameter of erythrocytes can increase or decrease in pathological cases.

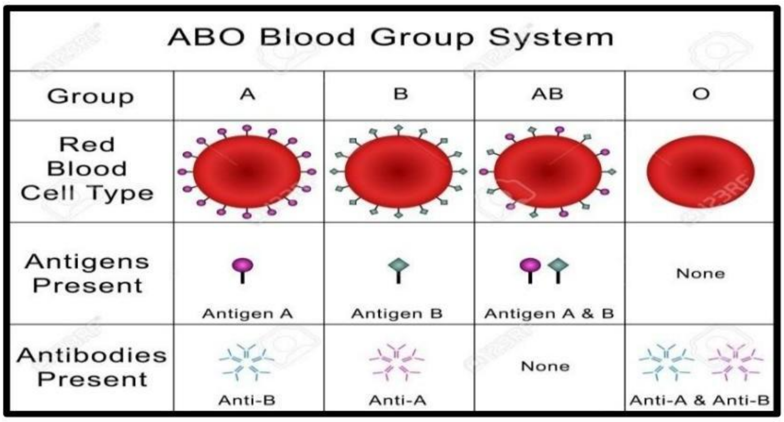
Microcytosis is characterized by red blood cells that are smaller than normal and is observed mainly in course of iron deficiency anemias and thalassemias.

Macrocytosis refers to red blood cells that are larger than normal. It is observed in cases of abnormal synthesis due to deficiencies in folate or vitamin B12.

- Normocytes are flexible cells, which allows them to adapt to the size of the vessels.

- If we place the RBCs in a hypotonic environment, they swell, they lose their biconcave disc shape, the plasma membrane will burst: this is hemolysis.
- Oxygen and carbon dioxide are transported via hemoglobin.
- Hemoglobin is made up of globin, a protein associated with four heme groups. Each heme group is associated with an iron atom
- Oxyhemoglobin is an unstable compound, which allows for a faster and easier exchange of oxygen at the tissue level.
-The combination of hemoglobin with CO_2 is much more stable.
-The set of phenomena leading to the formation of red blood cells is called erythropoiesis, which takes place in the bone marrow.
Anemia is a decrease in the concentration of red blood cells (RBCs) in the blood. Polycythemia is an increase in the concentration of RBCs in the blood.
-Polycythemia can be pathological or physiological.





ABO Blood Group System

Group	A	B	AB	O
Red Blood Cell Type				
Antigens Present	Antigen A	Antigen B	Antigen A & B	None
Antibodies Present	Anti-B	Anti-A	None	Anti-A & Anti-B

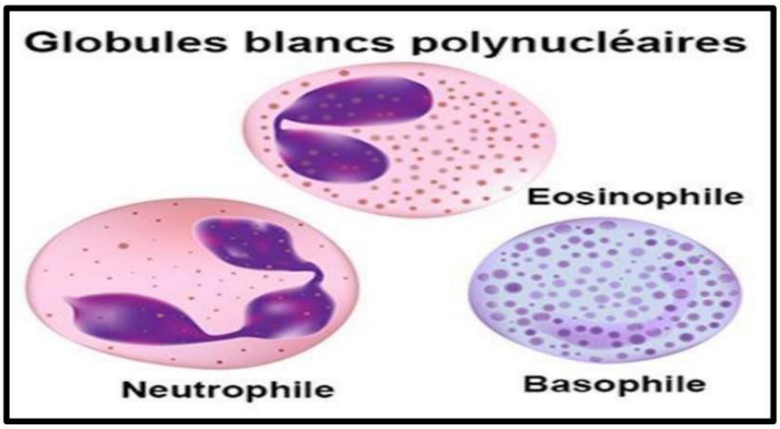
Fig. Erythrocytes, or Normocytes

<u>3-2- White blood cells, or leukocytes:</u>

- There are 4000 to 10000 of them per mm3 of blood, their nucleus may have many lobules and granules (polymorphonuclear = granulocytes) or one lobe and no granules (mononuclear = agranulocytes).
- Leukocytes are synthesized in the bone marrow and can pass into lymphoid organs, thus acquiring their specificity (lymphoid lineage), others go directly into the blood (myeloid lineage).

Their role is humoral (immunoglobulin) and cellular defense
- These are cells capable of amaeboid movements, and diapedesis (crossing capillaries by passing between two endothelial cells).

Fig 2: White blood cells under optical microscopy



A-Polymorphonuclear granulocytes

Neutrophils, 60 to 70% of WBCs

- Cells with a single nucleus and multiple lobes (2 to 5, depending on age), - A diameter of 12 microns

- The cytoplasm has 50 to 200 secondary bactericidal granules distributed regularly in the cytoplasm, appear dense under the EM and contain lysosomal enzymes (peroxidases, lysosomal enzymes and oxidases, alkaline phosphatase and phagocytin).

- These cells have a limited lifespan; they are terminal cells.

- They have a limited capacity for synthesis. They serve as the first line of defense, and are capable of phagocytosis.

- Neutrophils are metabolically active cells . **Eosinophils:** (limited number: 2 to 4% of WBCs).

- They have a diameter of 12 to 15 microns, and act in allergic phenomena.
- Rich in glycogen granules in the cytoplasm.
- The nucleus can be bi-lobed or tri-lobed.
- The granules of these cells are larger in size than in neutrophils.
- The granules contain ribonucleases and alkaline phosphatase.
- Eosinophils are capable of phagocytosis (amaeboid movements).
- These cells produce substances that have a neutralizing effect on leukotrienes and histamine.

Basophiles.

- The cells constitute 1% of the leukocytes
- They have a clover-shaped core.
- Their cytoplasm contains large basophilic granules; they secrete histamine (vasodilation), heparin (coagulation), and leukotrienes, which have an action in the slow contraction of a smooth muscle.

- In response to certain antigens, basophils release some of their granules
- These cells exert their action at the level of the tissues, they pass very quickly into these tissues, they are then called mast cells.
- Reactive in immediate hypersensitivity, they are also capable of amaeboid movement and phagocytosis.

B-Agranulocytes: hyaline leukocytes:

-Lymphocytes : 25 to 30% of WBCs, these are spherical cells of varying sizes: small, medium and large.

The small lymphocyte has a diameter of 6 to 8 microns, it has a nucleus with condensed chromatin, the nucleus is unique and spherical, it occupies the entire surface of the cell, the cytoplasm is very basophilic, it contains granules in small quantities, of azurophilic type.
- These cells are terminal and are capable of transforming in the presence of PHA (phytohemagglutinin), they will then become lymphoblasts (possibility of mitosis).

Large lymphocytes: have a more abundant cytoplasm, with a large number of organelles.

Intermediate lymphocytes
- They constitute a heterogeneous population, varying in size and morphology.
- They are also distinguishable.

Plasma cells : ovoid cells, 8 to 20 microns in diameter, with an eccentric, round nucleus and rough cytoplasm, involved in immunoglobulin production. Production occurs after antigenic stimulation.

Two major types are distinguished: B lymphocytes (immunoglobulin production) and T lymphocytes (cellular immune defense).

-Monocytes : 4 to 8% of WBCs.
Represents the undifferentiated, immature form of the macrophage.
It is the largest cell in the blood
- The nucleus is oval, it is said to be kidney-shaped, in an eccentric position, the chromatin is not very condensed, this nucleus has a fibrillar appearance, we speak of combed chromatin.
- The cytoplasm is slate-blue grey in color and contains azurophilic granules (lysosomes).
- The cells are able to cross the endothelial barrier, at which point they are called macrophages.

- Macrophages have an important macrophagocytosis function in tissues. This is the mononuclear phagocyte system.
- There are specific receptor sites that have a role in recognizing and interacting with the antigen.

Monocyte.
Lymphocyte.

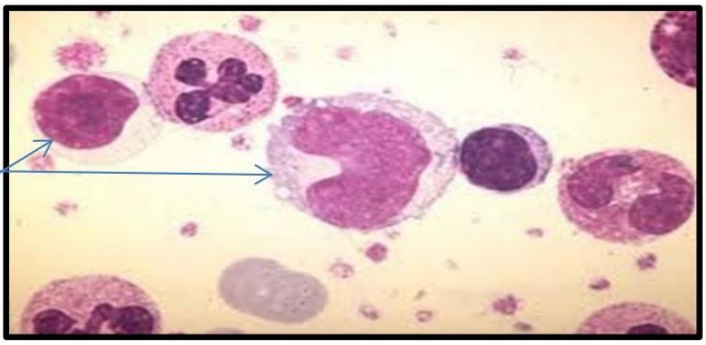


FIG 3 Monocyte under optical microscopy

3-3-Platelets, or thrombocytes;
- Blood platelets or thrombocytes are anucleate cell fragments 2 to 5 ÿm in diameter containing various granules and surrounded by a membrane - They originate from a giant cell, the megakaryocyte.
- Their lifespan is ten days; tubules and vesicles are observed in their cytoplasm. These tubules are formed by an invagination of the plasma membrane into the cytoplasm.

When there is an endothelial rupture, the blood contents will come into contact with the underlying connective tissue, in which there is collagen.
The platelets will rapidly adhere to the collagen via the collagen binding protein.

They will form the hemostatic plug that will fill the breach that will form at the level of the endothelial wall.

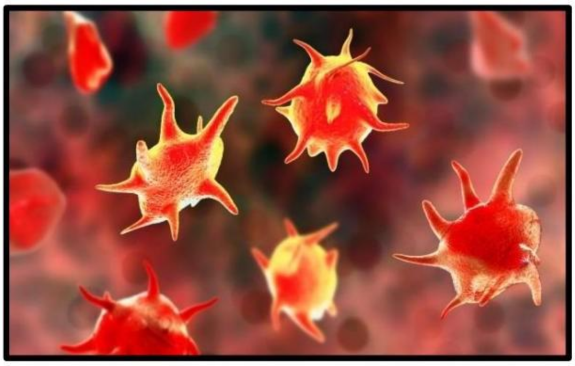


FIG 4: Thrombocytes

4- Role of White Blood Cells

Neutrophils: Their main role is the defense of the body by phagocytosis of foreign bodies, particularly bacteria.
Eosinophilic polymorphonuclear leukocytes: They destroy certain parasites by phagocytosis, and also suppress antigen-antibody complexes after a specific immune response.
Basophils: Their role is to participate in immuno-allergic reactions by releasing histamine contained in their granules.
Lymphocytes: They provide specific humoral and cellular defenses.
Monocytes: These are the precursors of macrophages that participate in non-specific and specific defenses.

5- Role of Platelets

- Platelets play a fundamental role in primary hemostasis and in inflammatory reactions where they activate chemotactic factors of polymorphonuclear leukocytes and macrophages.

6- Formation of the formed elements of blood: (Hematopoiesis)

The formed elements of blood have limited lifespans; there is a dynamic balance between their production (hematopoiesis) and their destruction.

Hematopoiesis

Hematopoiesis is a physiological, dynamic, continuous and regulated phenomenon ensuring the production of blood precursors (proliferation, differentiation and maturation) which takes place in hematopoietic organs (bone marrow in adults, liver and spleen in embryos).

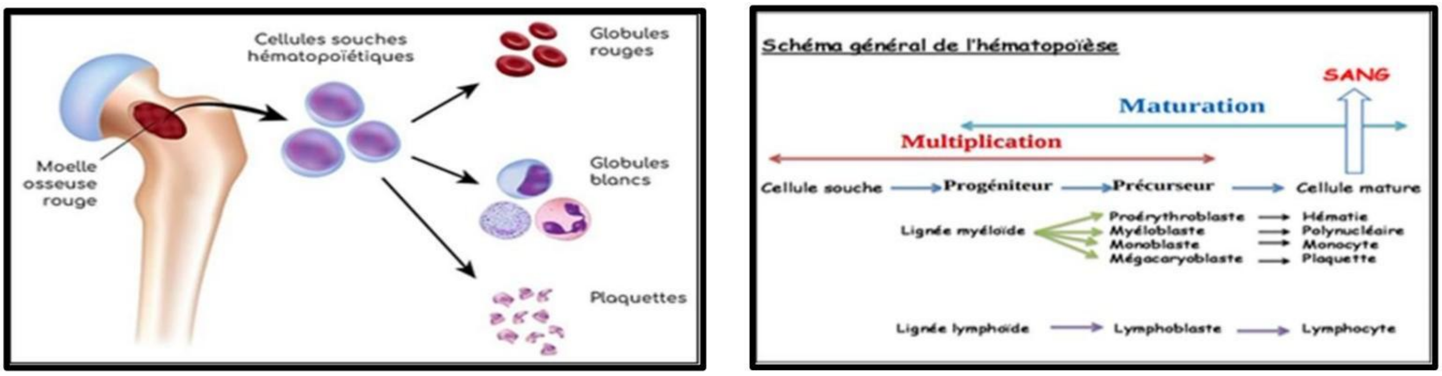


FIG 5: Hematopoiesis

Starting from a multipotent stem cell, two main differentiation pathways emerge:

-The lymphoid line and the myeloid line **-In a normal adult subject, only mature elements pass into the peripheral blood.**

6-1-Lymphoid Lineage:

-Lymphoblast: - 15 – 20 Um -

Nuclei with poorly stainable chromatin rays.

- Basophilic cytoplasm.

Prolymphocyte: - 15 µm

- Dense nuclei, 01 single nucleolus.
- Slightly basophilic cytoplasm.

- Lymphocyte: - cell of variable size 08 to 12 µm hence the name large, medium, small lymphocyte.

Note:

Lymphocytes that originate in the bone marrow are subdivided into two groups. Those that undergo an intrathymic stage before passing into the peripheral lymphatic organs (T lymphocytes) and those that pass directly into these organs (B lymphocytes).

6-2-Myeloid lineage:

These are cell lines specific to the bone marrow, 03 subunits: (erythrocyte, granulocytic, thrombocytic)

The erythroid lineage: normocytic.

Proerythroblast: - 20 – 25 Um

- Large nuclei with clear chromatin, nucleolated.
- Basophilic cytoplasm.

Erythroblast: - 16 – 18 Um -

Small nuclei with clumped or wheel-like chromatin.

- Basophilic cytoplasm.

Polychromatophilic erythroblast: - 09 – 10 Um -

Dark nucleus with dense chromatin,
its clump-like arrangement gradually disappears.

- Cytoplasm becomes enriched in Hemoglobin and turns dark pink.

Acidophilic erythroblast: - 8 – 10 Um

- Small, dense, very dark kernels.
- Pink-stained cytoplasm.

Reticulocytes: after expulsion of the nucleus, the acidophilic erythrocyte transforms into a reticulocyte: - Nuclei 4 – 7 Um.

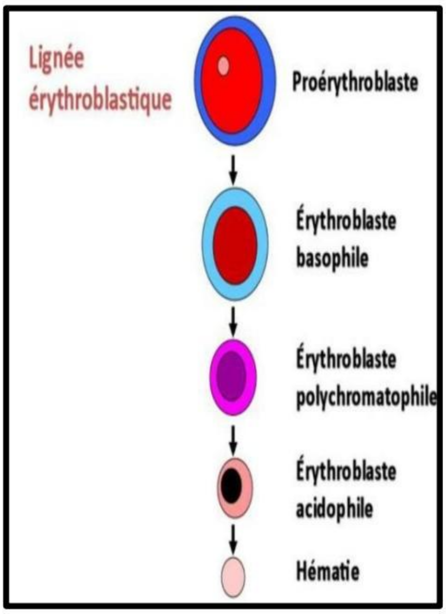


FIG6 The erythrocyte lineage

- Anucleate cell containing a reticulofilamentous substance , Being free, it gains the light of a capillary through the fenestrated wall of the medullary sinusoids and passes into the bloodstream.

- It is in the blood that 24 hours later it will become a mature red blood cell after having lost its movement capabilities and its organelles (this is the circulating form).

Granular cell lines:

The first identifiable granulocytic cell is the myeloblast; after 4-5 mitoses and following maturation phenomena dominated by the elaboration of specific granules, the myeloblast results in the polynuclear cell.

Myeloblast: - 20 – 25 µm
- Large nuclei with fine chromatin, and 4 to 5 nucleoli.
- Basophilic cytoplasm with some azurophilic granules.

Promyelocyte: - 18 – 20 µm
- Small nuclei, often eccentric and without a nucleolus.
- Cytoplasm less basophilic containing in addition to azurophilic granules specific granules hence the classic distinction promyelocyte N- AC or basophil.

Myelocyte: - 15 – 20 µm
- Nuclei small to large blocks of chromatin.
- Granulations.

Meta myelocyte: - 12 – 14 Um.
- Kidney-shaped nuclei.
- Granular cytoplasm.

Granulocyte: - 10 – 12 µm
- Lobed nuclei.
- Granular cytoplasm.

Thrombocyte lineage:

Megakaryoblast: - 20 – 30 Um
- Round kernels.
- Low-abundance basophilic cytoplasm.

Basophilic megakaryocyte: - 30- 50 Um
- Nx lobed.
- Basophilic cytoplasm.

Granular megakaryocyte: - 50-100 µm
- Multilobed nerve.
- Granular cytoplasm.

Thrombogenic megakaryocyte: - greater than 100 Um.
- Nxpicnotic then degenerates in the bone marrow.
- The cytoplasm fragments to form platelets.

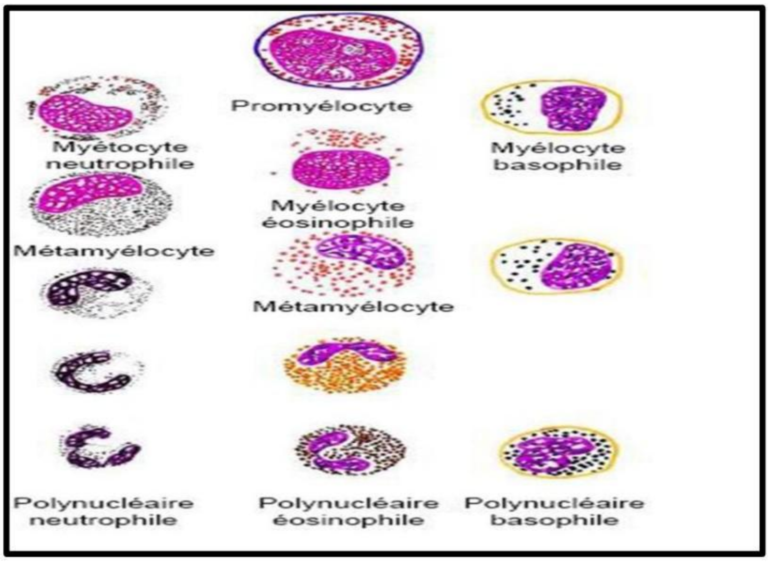


FIG 7: Granular lines

Monocytic lineage:

Monoblast: - 25 – 30 µm
- Oval-shaped nuclei possessing chromatin arranged in a fine striation.
- weakly basophilic cytoplasm.

Promonocyte: - 20 µm
- kernels comparable to the previous one.

- More differentiated cytoplasm.

Monocyte: - 15 – 25 Um -

Irregular nuclei, sometimes kidney-shaped.

- Cytoplasm possessing azurophilic granules and vacuoles.

7-Complete Blood Count: CBC, it measures the quantity of different cells per mm3 of blood.
The normal results are:

Numeration:

Red blood cells: 4500,000 to 5000,000 / mm3.

Leukocytes: 4000 to 10000 / mm3.

Platelets: 200,000 to 300,000 / mm3.

Formula:

Granular leukocytes: 60% to 75%.

Neutrophil granulocytes: 55% to 73%.

Eosinophilic granulocytes: 1.5% to 4%.

Basophilic granulocytes: 0.5%.

Hyaline leukocytes: 25% to 40%.

Lymphocytes: 20% to 30%.

Monocytes: 5% to 10%.